

Vinh Ly

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Summary

Software Engineer with **18 years** of experience designing and scaling distributed systems at Google, YouTube, and Whatnot. Built end-to-end infrastructure serving billions of requests daily, including video streaming pipelines at YouTube and real-time marketplace systems at Whatnot. Led teams of 8 engineers across search ranking, payments, and trust and safety domains. Built fault-tolerant services processing 10M+ daily transactions with sub-50ms latency at Google. Bringing deep full-stack expertise in **Python**, **React**, and **TypeScript** to a team building high-scale consumer applications. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages: Python, Go, Java, TypeScript, C++, JavaScript

Frontend: React, TypeScript, HTML/CSS, responsive design, component libraries

Backend: FastAPI, Python, REST APIs, microservices, GraphQL, caching strategies

Data: PostgreSQL, MariaDB, distributed databases, data modeling, query optimization

Infrastructure: GCP, Kubernetes, distributed systems, on-call rotation, SRE practices, Docker

Work Experience

Staff Software Engineer, YouTube

02/2025 - Present | Remote

- Migrated 50+ microservices to **Kubernetes** and modernized the API layer with **Python** and **FastAPI**, cutting average deployment time from hours to under 10 minutes.
- Owned video streaming infrastructure serving over a billion users with adaptive bitrate delivery, reducing playback failures by 40% through CDN optimization and failover systems.
- Led a team of 5 engineers maintaining infrastructure that sustained 100K+ concurrent connections with 99.99% availability and sub-50ms latency for live streaming.

Staff Software Engineer, Whatnot

05/2024 - 02/2025 | Remote

- Designed real-time marketplace infrastructure supporting 100K+ concurrent users with WebSocket connections for live auctions, handling 10K+ concurrent auctions with sub-100ms p99 latency.
- Built backend services with **Python** and **FastAPI** for bidding, checkout, and seller onboarding, achieving 30% improvement in API response times and increasing checkout completion rates by 15%.
- Developed React and TypeScript components for the seller dashboard and buyer experience, reducing page load times by 50% through lazy loading and code splitting.
- Owned database schema design for **PostgreSQL** and introduced caching with Redis, cutting database load by 60% and improving p99 latency from 200ms to under 80ms.

Staff Software Engineer/Manager, Google

03/2015 - 04/2024 | Remote

- Managed a team of 8 engineers building trust and safety systems that reduced fraudulent transactions by 30% through ML classifiers and rule-based detection systems.
- Owned search ranking infrastructure serving 1B+ daily queries, improving p50 ranking latency by 30% through caching and query optimization in **Python**.
- Built payments infrastructure processing \$1M+ daily transactions with 99.99% availability, eliminating payment failures through idempotency keys and retry logic in **Go**.
- Mentored 3 engineers who were promoted to senior engineer and established team norms for code review, on-call rotation, and incident response.

Software Engineer, Google

02/2011 - 02/2015 | Remote

- Built internal tooling in **Python** and **Java** to analyze search quality metrics and automate A/B test reporting, reducing manual effort by 80%.
- Participated in on-call rotation for production systems with strict SLOs, triaging incidents and conducting blameless postmortems to prevent recurrence.
- Developed data pipelines processing 10TB+ daily for indexing and ranking features using **distributed systems** on **GCP**.

Software Design Engineer II, Microsoft

10/2007 - 12/2011 | Remote

- Designed and implemented enterprise software components in **C++** and **C#** for a team with 500K+ internal users, working across the full software development lifecycle.
- Authored automated tests that increased code coverage from 50% to 80% and caught 20% more regressions before deployment.

Software Design Engineer Intern, Microsoft

06/2006 - 08/2006 | Remote

- Developed and tested internal web applications using **C#** and .NET, collaborating with full-time engineers on feature design and code reviews.

Education

B.S. Honors in Computer Science

Virginia Tech | 2004 - 2007